LAM: Learning Adaptation Model – Product Requirements Document

Oct 18, 2026 · @Mr.Forever

LAM is an independent, versioned service that turns structured student-learning evidence into a learner state and a recommended next teaching action (v0.1, draft).

1. Overview and goals

LAM is not a chatbot. Qrie and other learning apps consume it through a stable API, and it stays separate from Qrie's frontend, classroom logic and LLM provider.

Problem: tutors often cannot tell what a student understands, why they struggle, or whether an intervention worked.

Goals

- Estimate per-concept mastery, separating assisted from unassisted performance.
- Diagnose likely causes of errors, or ask a diagnostic question when evidence is weak.
- Recommend the next teaching action and schedule retrieval reviews.
- Track which strategies work, with uncertainty.
- Serve any LMS through a standard API and SDK.

Primary metric: unassisted mastery gain per learning minute.

Supporting metrics: delayed retention, transfer, and recommendation acceptance rate.

Non-goals: generating lessons or chat replies, authentication, billing, school administration, fixed learner-type labels, and training on raw conversations.

Consumers: the Qrie classroom and tutor (recommendations and state), third-party LMS apps (REST and SDK), learners and teachers (dashboards via the host app), and the LAM team (audit logs, evaluation, rollout).

2. Scope by release

The immediate target is M0 and M1. Neural-network training waits until measured evidence shows it beats a simpler baseline.

Release	Focus	Contents
M0	Service foundation	Repo scaffold, typed schemas, auth, PostgreSQL migrations, versioned API, event ledger, tests, SDK skeleton
M1	Deterministic learner model	Per-concept state, BKT mastery, assisted/unassisted tracking, misconception evidence, rule-based next action
M2	Closed loop with Qrie	Diagnosis, adaptive teaching, independent assessments, delayed retrieval, persistent state
M3	Personalization	Strategy effectiveness, uncertainty calibration, policy evaluation
M4	Shared intelligence	Approved multi-student data, offline evaluation, model versioning, controlled rollout and rollback

3. Architecture

A request flows through six stages: API gateway and auth, evidence validation, diagnosis engine, learner state engine, adaptive decision engine, and recommendation API.

Private data layer: student profiles, learning event ledger, curriculum/concept graph (consumed through a stable interface), and strategy effectiveness.

Shared intelligence (separate and governed): approved dataset, then training and evaluation, then a versioned shared model. Training is never an automatic side effect of API requests, and the shared model never stores identifiable histories.

Responsibility split

- Shared model: general patterns across students.
- Student profile: one learner's mastery, misconceptions, retention and strategy outcomes.
- Decision engine: combines both with curriculum constraints and the current context.
- LLM: external and replaceable, invoked by the consuming tutor when the decision engine requests it.

4. Input and output contracts

LAM accepts structured evidence, not raw transcripts. A missing signal means "unknown", never "negative".

Input group	Fields	Status
Identity and context	`event_id`, `student_id`, `concept_id`, `item_id`, `timestamp`	Required (MVP)
Response	`correctness`, `attempt_number`	Required (MVP)
Question	`difficulty`, `concept_ids`	Recommended
Assistance	`hints_used`, `assistance_level`	Recommended
Timing and confidence	`response_time_ms`, `self_reported_confidence`	Recommended
Teaching context	`strategy_id`, `intervention_id`	Recommended
Curriculum	`subject`, `grade`, prerequisite graph reference	Recommended
Historical state	Learner-state version or server-side profile reference	Recommended

Identity must come from trusted credentials, not a client-supplied `student_id`. Working steps, voice, handwriting and engagement signals arrive in later releases, each with its own privacy controls.

Output

- `learner_state`: mastery estimate, unassisted mastery, assisted performance, uncertainty
- `diagnosis`: candidate causes with confidence, plus `needs_confirmation`
- `next_action`: type, strategy, assistance level, target concept
- `review`: required flag and recommended interval
- A short rationale (no chain-of-thought) and `state_version`

Scores need explicit definitions and must not imply scientific precision until calibrated and validated.

Example input (illustrative)

```
{
  "event_id": "evt_1042",
  "student_id": "student_123",
  "concept_id": "fractions.unlike_denominators",
  "item_id": "q_892",
  "timestamp": "2026-10-18T14:30:00Z",
  "correctness": false,
  "attempt_number": 2,
  "hints_used": 1,
  "assistance_level": 1,
  "response_time_ms": 42000,
  "self_reported_confidence": 0.8,
  "strategy_id": "visual_explanation_v1"
}
```

Example output (illustrative)

```
{
  "student_id": "student_123",
  "concept_id": "fractions.unlike_denominators",
  "learner_state": {
    "mastery_estimate": 0.55,
    "unassisted_mastery": 0.42,
    "assisted_performance": 0.78,
    "uncertainty": 0.30
  },
  "diagnosis": {
    "candidates": [
      {"type": "conceptual_misunderstanding", "label": "Denominator
addition", "confidence": 0.65}
    ],
    "needs_confirmation": true
  },
  "next_action": {
    "type": "ask_diagnostic_question",
    "strategy": "socratic_question",
    "assistance_level": 2,
    "target_concept": "fractions.unlike_denominators"
  },
  "review": {"required": true, "recommended_interval_hours": 48},
  "state_version": 12
}
```

5. Functional requirements

ID	Requirement	Description
FR-01	Learner state	Versioned, auditable per-concept mastery, uncertainty, misconceptions, assisted/unassisted performance, retention
FR-02	Diagnosis	Candidate causes with confidence; ask a diagnostic question when evidence is insufficient
FR-03	Adaptive decisions	Choose among independent practice, metacognitive prompts, hints, analogies, worked examples, prerequisite remediation, direct explanation
FR-04	Mastery verification	Confirm mastery only with independent assessment evidence
FR-05	Sequencing	Use mastery, prerequisite gaps, misconception severity, forgetting risk, curriculum importance, difficulty
FR-06	Retrieval scheduling	Track last practiced, history, memory strength and next review; update on success or failure
FR-07	Strategy effectiveness	Outcomes per concept and learner, accounting for sample size, uncertainty and confounders
FR-08	Model lifecycle	Versioned training, offline evaluation, rollout, monitoring, rollback
FR-09	Explainability	Action, rationale, evidence and uncertainty; no internal chain-of-thought
FR-10	Integration	Versioned APIs, SDKs, idempotent event submission, state retrieval, schema migration policy

6. Non-functional requirements

The service must be independently deployable, testable and replaceable without changing Qrie's core classroom architecture.

- **Isolation:** server-side authorization and tenant-aware access on every endpoint. A student ID in the URL is not authorization.

- **Privacy:** data minimization, defined consent, retention, deletion and training eligibility.
- **Reliability:** idempotent ingestion, concurrency-safe state updates, recoverable failures.
- **Performance:** latency targets set from real classroom needs; training stays off the request path.
- **Versioning:** API schemas, learner-state schema, decision policies and shared models versioned independently.
- **Observability:** request IDs, model versions, decision reasons, errors and latency, without logging sensitive student content.
- **Portability:** HTTP/JSON, a documented Python SDK and a TypeScript client.
- **Safety:** validate recommendations; escalate on high uncertainty or repeated struggle.

7. API surface and tech stack

The v1 API is small and versioned. Every endpoint validates the caller's permission to access the learner.

Endpoint	Purpose
`POST /v1/events`	Ingest a learning event and update state
`GET /v1/students/{id}/state`	Retrieve the authorized learner profile
`POST /v1/students/{id}/diagnose`	Evaluate a suspected misconception or knowledge gap
`POST /v1/students/{id}/recommendations`	Select the next best learning action
`POST /v1/students/{id}/assessments`	Record assessment evidence
`GET /v1/students/{id}/reviews`	Retrieve due or upcoming retrieval tasks
`GET /v1/health`	Check service health
`GET /v1/models/current`	Report the active shared model version

Proposed SDK interface (not yet existing):

```
lam.record_event(event)
lam.get_learner_state(student_id)
lam.recommend_next_action(student_id, concept_id)
lam.record_assessment(result)
lam.get_due_reviews(student_id)
```

Layer	Choice	Reason
API	FastAPI + Pydantic	Typed contracts, easy integration
Decision engine	Python rules and deterministic policies	Easy to test, explain and debug
Mastery model	Bayesian Knowledge Tracing (or another validated approach)	Models knowledge updates from interaction evidence
Concept graph	Consume Qrie's graph via a stable interface	Avoid duplicating curriculum infrastructure
Storage	PostgreSQL	Transactional state updates and event records
Background jobs	Task queue only when volume requires it	Keep heavy processing off the request path
ML	scikit-learn first, PyTorch only when justified	Avoid premature complexity
Model tracking	Versioned artifacts and evaluation reports	Reproducible deployment and rollback
Integration	REST + Python SDK + TypeScript client	Works with Qrie and other apps
Deployment	Docker + CI	Reproducible, independent releases

BKT is a reasonable starting point for concept-level mastery but is not a complete diagnosis or teaching-strategy model. Effectiveness must be evaluated, not assumed.

8. Boundaries, acceptance criteria and risks

What LAM must not do

- Generate every lesson or reply itself.
- Treat every wrong answer as a misconception.
- Assign permanent learner types (visual, auditory, etc.).
- Count assisted success as independent mastery.
- Train on all raw conversations, or update global weights after every answer.
- Infer fatigue, boredom or emotion with unjustified confidence.
- Own authentication, billing or school administration.

Acceptance criteria (v1)

1. A retried event updates the correct concept state exactly once.
2. A likely misconception triggers a diagnostic question or intervention.
3. Assisted and unassisted mastery are stored separately.
4. A later independent assessment can change mastery and the review schedule.
5. Recommendations respect prerequisites.
6. Student A cannot read or modify Student B's profile.
7. Qrie can use the API without importing LAM's internal packages.
8. Models can be evaluated and rolled back without losing learner history.
9. Decisions and state updates are reproducible in known test scenarios.
10. Evaluation shows improvement on a predefined outcome against a baseline before any effectiveness claim.

Risks: BKT misfit on some concepts, confounded strategy-effectiveness data, privacy and consent for shared training, and over-trusting uncalibrated scores.

Open questions

- Latency targets
- Single-tenant vs multi-tenant
- Concept-graph interface format
- Consent model for M4 training data
- Baseline for the effectiveness evaluation

Ownership boundary: LAM owns the learner model, evidence processing, diagnosis, adaptive decisions, review scheduling and the shared-model lifecycle. Qrie owns the student UI, content delivery, lesson rendering, tutor conversations, voice and school workflows. They communicate only through versioned contracts.

9. Gap audit and extended requirements

Sections 1 to 8 specify the service, its contracts and its boundaries. Sections 10 to 20 below turn the underspecified capabilities into concrete requirements. Every default value in them (thresholds, intervals, counts) is a proposed starting point to be tuned against data, not a validated constant.

Capability	Status in sections 1 to 8	Specified in
Observe, diagnose, teach, test, update, revisit loop	Partial	Section 10
Persistent learner profile	Partial	Section 11
Productive-struggle assistance ladder	Partial	Section 12
Misconception diagnosis	Partial	Section 13
Adaptive sequencing	Partial	Section 14
Curriculum knowledge graph	Partial	Section 15
Spaced retrieval and forgetting	Partial	Section 16
Teaching-strategy effectiveness	Partial	Section 17
Learning Scientist Agent	Missing	Section 18
Answer verification	Partial	Section 19
Privacy, child safety, governance	Partial	Section 20
Proactive intervention, metacognition, engagement signals	Partial	Section 20
Multimodal evidence	Deferred	Section 20
Assisted vs unassisted mastery	Covered, needs detail	Sections 11 and 19
Measurement of learning effectiveness	Partial	Sections 17 and 18

10. Closed-loop learning engine

LAM is a continuous process, not a request handler that maps one correct/incorrect event to one recommendation. Each learning session follows a defined lifecycle, and every stage feeds the next.

Lifecycle stages

1. **Observe:** collect attempts, hints, timing, confidence and intervention context from Qrie.
2. **Diagnose:** estimate plausible causes of the error and record how uncertain they are.

3. **Select intervention:** choose the next teaching action and assistance level.
4. **Deliver and record:** Qrie delivers the intervention and reports what actually happened, which may differ from what LAM recommended.
5. **Verify learning:** use an independent, unassisted assessment when the policy calls for one.
6. **Update learner state:** revise mastery, misconceptions, retention and strategy evidence.
7. **Schedule next action:** continue, remediate a prerequisite, or schedule a retrieval review. The result produces new evidence and re-enters stage 1.

Session state machine

Each student-concept pair in an active session is in exactly one state. LAM persists the state with the learner-state version so a retried or resumed session continues from the same point.

State	Entered when	Exits to	Exit rule
`observing`	A session starts or a new event arrives	`diagnosing`	An incorrect or assisted event, or a proactive trigger fires
`observing`	Correct, unassisted event	`verifying` or `scheduling`	Verification due, else schedule next action
`diagnosing`	Evidence suggests a cause is needed	`intervening`	Top hypothesis confidence meets threshold, or a diagnostic question has been chosen
`intervening`	An action is recommended	`awaiting_result`	Qrie confirms delivery
`awaiting_result`	Intervention delivered	`observing`	Next event received; timeout returns to `scheduling`
`verifying`	Independent assessment due	`updating`	Assessment result recorded

State	Entered when	Exits to	Exit rule
`updating`	New evidence ready	`scheduling`	State version incremented successfully
`scheduling`	State updated	`observing` or `closed`	Next activity or review chosen; or the session ends
`escalated`	Repeated failure, stop rule, or no safe recommendation	`observing` or `closed`	Human or host-app action resumes or ends it

Required transition behavior

- **Incorrect answer with strong reasoning:** do not treat as a gap. Mark as a probable slip or procedural error, lower the misconception hypothesis weight, and prefer a short independent retry.
- **Correct after a hint:** record as assisted success. It raises assisted performance but contributes little or nothing to independent mastery.
- **Repeated failure after progressively stronger help:** stop repeating the same level. Move to a diagnostic question, a prerequisite check, or `escalated` per the assistance policy in section 12.
- **Diagnostic question fails to separate two hypotheses:** keep both with their confidences, select a different discriminating question, and cap the number of diagnostic attempts per concept per session.
- **Mastery shown in-lesson but delayed review failed:** reduce the durable-retention estimate, shorten the next interval, and flag the concept as fragile rather than mastered.
- **Evidence or curriculum graph unavailable:** return a safe fallback (continue current activity, or defer to the host app) with an explicit `degraded` flag and reason. Never invent a recommendation.

Terminal conditions and recovery

A session ends in `closed` when the student stops, the host app ends it, or a goal is reached. Events remain idempotent, so a crash during `updating` is recovered by replaying the ledger from the last committed state version. Retries use the same `event_id` and produce the same result.

11. Learner state data contract

The learner state is a set of separate dimensions, never one opaque score. Each dimension is stored per student and per concept, with its own uncertainty, evidence count and last-updated time.

Dimension	Meaning	Representation (proposed)	Updated by
Concept mastery	Estimated knowledge of a concept	Probability in [0,1] from the knowledge-tracing model	All scored events, weighted by assistance and evidence quality
Independent mastery	Success without assistance	Probability in [0,1], plus count of independent attempts	Unassisted attempts and independent assessments only
Assisted performance	Performance with hints or explanations	Success rate in [0,1] by assistance level	Assisted attempts
Misconception state	Candidate misconception with status	List of hypotheses: confidence, supporting and contradicting evidence IDs, status (`open`, `confirmed`, `retired`)	Diagnosis engine
Prerequisite readiness	Whether required concepts are sufficiently understood	Per-prerequisite flag plus the mastery value behind it	Mastery of prerequisite concepts in the graph
Retention	Persistence over time	Memory-strength estimate, last recall time, delayed-recall history	Reviews and delayed assessments

Dimension	Meaning	Representation (proposed)	Updated by
Confidence calibration	Self-reported confidence vs actual outcome	Calibration gap over a rolling window of events with confidence	Events that include `self_reported_confidence`
Strategy effectiveness	Outcomes after a teaching strategy	Per strategy: n, outcome estimate, interval (section 17)	Post-intervention assessments
Evidence quality	How reliable the observations are	Recency, count, independence, difficulty, verifier reliability	Every event
Uncertainty	How unsure the system is	Width of the credible interval or an equivalent, per estimate	Every update

Semantics rules

- Every score must state whether it is a probability, an index or a count. A mastery value of 0.55 is a model estimate under stated assumptions, not a measured percentage of knowledge, until calibration shows otherwise.
- Uncertainty is reported alongside every estimate. A high-uncertainty estimate must not trigger a progression or mastery claim.
- A missing signal is stored as unknown and excluded from the update, never as zero.
- Mastery is confirmed only when independent mastery passes the threshold in section 19 and the evidence is verified, not provisional.

Update and history rules

- Each accepted event produces one new state version. Versions are immutable and monotonically increasing per student.
- A snapshot stores the state, the event IDs applied, the model and policy versions, and a timestamp. Past snapshots are retained per the retention policy in section 20.
- An update carrying a stale `state_version` is rejected or merged with a documented rule. Concurrent updates are serialized per student.
- Calibration of scores is evaluated on held-out data before any score is shown to learners as a percentage.

12. Assistance ladder policy

LAM recommends how much help to give, in a deliberate order that preserves productive struggle. Qrie reports which level was delivered and the outcome, and LAM recommends the next level.

Level	Action	Entry condition	Escalate when	Success condition
0	Independent attempt	Default for any new item	Incorrect or abandoned, or long inactivity	Correct, unassisted
1	Prompt to rethink	First incorrect attempt	Second incorrect attempt	Correct on retry
2	Conceptual question	Rethink prompt did not help	Answer shows the concept is unclear	Student states the right idea
3	Small hint	Level 2 failed or diagnosis points to a specific step	Failed after the hint	Correct after hint (assisted)
4	Stronger hint	Small hint failed	Failed again	Correct after hint (assisted)
5	Similar worked example	Hints failed or prerequisite gap suspected	Failed on the following attempt	Correct on a parallel item
6	Partial solution	Worked example failed	Failed again	Student completes remaining steps
7	Full explanation	Partial solution failed	Student still fails	Correct on a new item
8	Direct answer	Only when justified (time limit, safety, repeated failure with a stop rule)	Not applicable	Returns to practice at level 0

Rules

- **Maximum repetition:** never recommend the same level more than twice in a row for the same item. After two failures at a level, escalate, switch to a diagnostic question, or address a prerequisite.

- **Skipping levels:** allowed when diagnosis is confident of a missing prerequisite or when the evidence shows the student has already tried the lower levels.
- **Stopping policy:** after a defined number of failed escalations (proposed: three levels without success), stop escalating. Offer a break, suggest teacher or parent help, or return to a simpler item. Mark the state `escalated`.
- **Return to independence:** after any assisted success, the next item on the concept is attempted at level 0. Only that outcome counts toward independent mastery.
- **Direct answer:** recorded as an assisted event. It never contributes to independent mastery.
- **Every recommendation** includes the level, the reason, and the previous levels and outcomes it considered.

13. Diagnosis as hypothesis testing

A wrong answer is evidence, not a verdict. The diagnosis engine keeps a set of competing hypotheses per student and concept, each with supporting and contradicting evidence, and reports uncertainty instead of forcing a label.

Taxonomy (initial)

Hypothesis	Typical evidence	Typical next step
Conceptual misunderstanding	Consistent wrong pattern across items; confident wrong answers	Targeted diagnostic question, then concept explanation
Missing prerequisite	Failure on items needing a prerequisite; prerequisite mastery low	Prerequisite check or remediation
Procedural or arithmetic error	Correct approach with a calculation mistake	Short retry, procedure practice
Misreading the question	Answer fits a different question; very short response time	Rephrase and retry
Guessing	Very fast answers, random pattern, low confidence	Lower-stakes check, ask for reasoning
Memory or retrieval failure	Previously correct, now incorrect after a gap	Retrieval practice, review schedule

Hypothesis	Typical evidence	Typical next step
Carelessness or isolated slip	Single error among correct answers	Independent retry
Incorrect reasoning strategy	Systematic but flawed method	Worked example contrasting methods

The taxonomy is versioned and extended per subject. Concept-specific misconceptions (for example, adding denominators) are registered under the conceptual hypothesis with their own identifiers.

Evidence model

- Each observation is attached to a hypothesis as supporting or contradicting, with a weight set by evidence quality (section 19).
- Confidence per hypothesis is recomputed on every relevant event and is never allowed to reach certainty from a single event.
- Hypotheses are reported as candidates with `needs_confirmation` true until confidence passes the confirmation threshold (proposed: 0.8) and at least two independent observations support it.

Hypothesis lifecycle

1. **Open:** created when evidence first supports it.
2. **Confirmed:** threshold met; interventions target it.
3. **Revised:** new evidence shifts weight to another hypothesis.
4. **Retired:** contradicted repeatedly, or the student demonstrates independent mastery after intervention.

Diagnostic questions

When two or more hypotheses are close in confidence, LAM recommends the question expected to best separate them and states which hypotheses it tests. If evidence is insufficient after the allowed diagnostic attempts, LAM reports the uncertainty explicitly and falls back to a general intervention at the next assistance level.

14. Adaptive sequencing decision policy

The first version is a deterministic, versioned, testable policy. It filters actions by hard constraints, ranks the eligible ones, resolves ties predictably, and records why the winner won.

Step 1: hard constraints (never violated)

- Do not introduce a concept whose prerequisites are below readiness, unless the action is remediating them.
- Do not count assisted evidence toward progression.
- Do not recommend content outside the learner's curriculum scope or grade range supplied by the host app.
- If the safety rules trigger (repeated struggle, stop rule), the action is `escalate` regardless of rank.

Step 2: eligible candidate actions

Independent practice, diagnostic question, hint at the next assistance level, worked example, prerequisite remediation, direct explanation, retrieval review, mastery assessment, or break.

Step 3: ranking factors

Priority	Factor	Effect
1	Open safety or stop condition	Forces escalation
2	Missing prerequisite blocking the current concept	Prefer prerequisite remediation
3	Confirmed misconception in the current concept	Prefer targeted intervention
4	Overdue retrieval review with high forgetting risk	Prefer review, unless it interrupts an active diagnosis
5	Mastery verification due	Prefer independent assessment
6	Curriculum importance and goal fit	Boost foundational concepts; exam goals may raise priority without overriding prerequisites
7	Difficulty fit	Select an item near the learner's current level

Conflict and tie rules

- Higher priority wins. Within a priority, prefer the action with higher expected learning value, then the older evidence, then the lower concept ID for determinism.
- Independent practice is chosen when independent mastery is uncertain and no blocking gap or misconception exists. Remediation is chosen when a prerequisite is below readiness or the same concept failed after the stop-escalation rule.

- Exam goals can reorder concepts within the allowed set but never skip foundational prerequisites.

Worked example

A learner has a weak prerequisite, an overdue review, and a misconception in the current lesson. The weak prerequisite blocks the current concept, so remediation wins. The review is queued with its due time preserved. The misconception hypothesis stays open and is addressed after the prerequisite check.

Explainability and test cases

Every recommendation returns the winning rule, the factors considered and the policy version. The policy ships with representative scenario tests (weak prerequisite plus overdue review, repeated failure, fresh success, skipped review) that must pass and remain reproducible for any change to the policy.

15. Concept graph contract

Qrie owns the curriculum. LAM consumes a versioned concept graph through a stable interface and never keeps an inconsistent second copy. LAM caches a read-only snapshot keyed by graph version.

Requirements

- **Stable concept IDs:** an ID never changes meaning. Renames keep the ID; splits and merges publish a mapping from old IDs to new ones.
- **Prerequisite edges:** directed edges from a concept to the concepts it requires, with an optional strength (`required` or `recommended`). The graph must be acyclic; LAM validates this on load and rejects a cyclic version.
- **Curriculum references:** each concept carries subject, grade, curriculum and syllabus references.
- **Question-to-concept mapping:** every item maps to one or more concepts, with a primary concept and optional weights for multi-concept items. Evidence from a multi-concept item is distributed by those weights.
- **Versions:** the graph has a version identifier. Every learner-state snapshot records the graph version it was computed against.
- **Missing data behavior:** an unknown concept returns a `graph_missing_concept` error and no state update. A missing prerequisite is treated as unknown readiness, which blocks progression and sets the `degraded` flag.
- **Compatibility:** Qrie must announce breaking graph changes ahead of publishing them. LAM supports the previous graph version for a defined overlap period.

Sample payload (illustrative)

```
{
  "graph_version": "2026.10.1",
  "concepts": [
    {
      "concept_id": "fractions.unlike_denominators",
      "subject": "math",
      "grade": 5,
      "curriculum_ref": "cbse.math.g5.fractions",
      "prerequisites": [
        {"concept_id": "fractions.equivalent", "strength": "required"},
        {"concept_id": "multiples.lcm", "strength": "recommended"}
      ]
    }
  ],
  "items": [
    {
      "item_id": "q_892",
      "primary_concept_id": "fractions.unlike_denominators",
      "concept_weights": {"fractions.unlike_denominators": 0.8,
"fractions.equivalent": 0.2},
      "difficulty": 0.6
    }
  ]
}
```

Validation rules: acyclic prerequisites, every referenced ID exists, weights per item sum to 1, difficulty in [0,1], and no concept lists itself as a prerequisite.

16. Retrieval scheduling algorithm

LAM starts with a documented baseline scheduler and evaluates it. It does not claim a precise forgetting curve until real data supports that model.

Baseline: expanding-interval ladder (proposed defaults)

Review outcome	Effect on next interval
First independent success after learning	Schedule at 1 day
Independent success at a review	Multiply the interval by 2 to 2.5, within a configured maximum

Review outcome	Effect on next interval
Success with assistance	Keep the interval unchanged or shorten it by half; do not count as durable retention
Failure	Reset to the shortest interval (1 day) and mark the concept fragile
Skipped review	Mark overdue, increase forgetting risk with time since due, and surface it with priority per section 14; do not count as success or failure

Interpreting evidence

- **Review difficulty:** success on a harder item extends the interval more than success on an easy item. Failure on an easy item lowers memory strength more.
- **Assistance during review:** an assisted success is partial evidence only.
- **Durable vs temporary performance:** a concept is durably retained only after at least two independent successes separated by a delay (proposed: at least one scheduled interval). A success in the same session as learning counts as temporary performance.
- **Repeated recall:** each successive independent success lengthens the interval until the configured maximum.

Persisted scheduling state (per student and concept)

- `last_practiced_at`, `last_review_outcome`, `review_count`, `consecutive_successes`
- `memory_strength` estimate and its uncertainty
- `interval_hours` and `next_review_at`
- `status`: `new`, `learning`, `review`, `fragile`, `overdue`
- `scheduler_version`

Acceptance tests

- Success, failure, skipped review and delayed-retention scenarios each produce the documented interval and status.
- A failed review after an in-session success lowers retention and shortens the next interval.
- The scheduler is deterministic for a given history and version.
- Scheduler variants are compared offline on delayed-retention outcomes before any change ships.

17. Strategy effectiveness methodology

A simple link between a strategy and a correct answer is not evidence of effectiveness. A worked example is often used because a learner is struggling, so a worse outcome afterward does not prove it is worse than a visual explanation used on an easier case.

Strategy-outcome record (per intervention)

- Learner and concept IDs, strategy ID and version, intervention ID
- Learner prior state at selection time: mastery, uncertainty, active hypotheses, assistance level, recent failures
- Reason the strategy was selected (policy rule or exploration)
- What Qrie actually delivered
- Outcomes, each timestamped: immediate response improvement, independent performance on the next unassisted item, delayed retention at the next review, and transfer to a new question or context

Outcome types kept separate

Outcome	Measures	Weight in effectiveness
Immediate improvement	Correct response shortly after the intervention	Low, since help is still fresh
Independent performance	Unassisted success afterward	High
Delayed retention	Success at the scheduled review	High
Transfer	Success on a new item or context	High

Estimation rules

- Compare strategies only among learners and situations with similar prior state (matched or stratified by prior mastery, concept and difficulty).
- Report an effect estimate with an interval and the sample size. Do not rank strategies below a minimum sample size (proposed: 30 comparable interventions per strategy and concept group).
- Label every estimate as `insufficient_evidence`, `suggestive` or `supported`, and let the decision policy use only the supported ones as tie-breakers.
- Begin with transparent, conservative estimates. Add controlled experimentation, such as randomized strategy selection among equally eligible options, only with an approved evaluation design and learner-safety review.

- Strategy effectiveness is stored per concept and per learner group. It is never turned into a fixed learner type such as visual or auditory.

Learning-effectiveness evaluation

Before any claim that LAM improves learning, run a controlled comparison against a predefined baseline (for example, fixed sequencing without adaptation) on unassisted mastery gain per learning minute, delayed retention and transfer. Set the minimum evidence standard (sample size, duration, success criterion) before the evaluation starts.

18. Learning Scientist subsystem

The Learning Scientist is a post-session evidence-analysis component, not an unconstrained autonomous agent. It never blocks ordinary event ingestion.

Responsibilities

- Collect structured evidence for a session.
- Summarize demonstrated strengths and weaknesses.
- Identify misconceptions that the evidence supports.
- Compare assisted performance with independent assessment results.
- Evaluate the outcomes of the teaching interventions delivered.
- Propose learner-state updates and future practice.
- Recommend retrieval reviews.
- Record the evidence IDs, policy versions and model versions behind every conclusion.

Inputs and outputs

	Content
Inputs	Session event ledger slice, interventions delivered, assessment results, current learner state, concept graph version
Outputs	Session summary, proposed state updates with confidence, misconception findings, recommended reviews, evidence references

Execution and permissions

- **Timing:** runs asynchronously when a session closes or goes idle, as a background job. It is not on the interactive request path.
- **Implementation order:** start as deterministic aggregation of session evidence. A model-assisted analysis may be added later behind the same output contract.

- **State-update permission:** the subsystem can propose updates. Low-impact, evidence-backed updates may be applied automatically. Uncertain or high-impact conclusions (such as confirming mastery or retiring a misconception) stay `provisional` until independent evidence validates them.
- **No silent overwrite:** an unverified narrative must never replace the authoritative learner state.

Failure handling

If the job fails, the session evidence stays in the ledger and the job is retried with idempotent processing. After the retry limit, the failure is logged with the session ID and an alert is raised. The learner state remains at its last committed version, and recommendations continue using it.

19. Answer verification and evidence provenance

A Boolean `correctness` field says what was observed, not how reliable the judgment is. LAM records how correctness was established and weights the evidence accordingly. Qrie stays responsible for delivering questions and running the right verifier, and sends the result to LAM through this contract.

Verification-result schema (proposed fields)

Field	Meaning
`assessment_id`, `item_id`	The assessment and item the evidence belongs to
`correctness`	Observed result, or `partial` with a score in [0,1]
`judgment_source`	`symbolic_math_check`, `code_tests`, `answer_key`, `curriculum_fact_check`, `teacher`, `model_graded`
`verifier_version`	Version of the verifier used
`verifier_reliability`	Known reliability class for this source (high, medium, low)
`independently_produced`	True only if no hint or explanation preceded the answer
`assistance_level`	Highest assistance level delivered before the answer
`evidence_for`	Mastery or misconception evidence this result supports, with IDs

Field	Meaning
`status`	`provisional` or `verified`

Rules for which evidence may update mastery

- Only `independently_produced` results with status `verified` count toward independent mastery.
- Model-graded or low-reliability results are stored as `provisional` and may influence diagnosis with reduced weight, but cannot confirm mastery.
- Provisional evidence becomes verified when a higher-reliability verifier confirms it, or when a later independent assessment agrees.
- Evidence weight scales with verifier reliability, item difficulty and recency.
- Results missing `judgment_source` are accepted as provisional with a warning, never silently trusted.

Mastery confirmation criteria (proposed defaults)

- Independent mastery estimate at or above 0.9 with narrow uncertainty.
- At least 3 independent, verified successes across at least 2 sessions, including one item at or above medium difficulty.
- No open confirmed misconception on the concept.
- Delayed retention check passed at least once before the concept is called durably mastered.

20. Privacy, child safety, governance and signals

Consent and training eligibility

- Collect only the fields needed for the contract in section 4.
- Each learner record carries consent flags for service use and for inclusion in shared-model training. Training eligibility defaults to excluded until consent is recorded by the host app.
- Shared-model datasets are approved, privacy-controlled extracts from the ledger. Identifiers are removed or pseudonymized, and a dataset approval is logged before any training job runs.
- Where learners are minors, consent comes from the guardian or school through the host app, and LAM stores only the consent reference and status.

Retention and deletion

- Retention schedules are set per data class: raw events, state snapshots, derived features, audit logs and training extracts.
- A deletion request removes the learner's profile, ledger events and snapshots, and flags any approved training extract for exclusion from future training. The request and its completion are logged without storing the deleted content.
- Deletion must not corrupt other learners' data or break reproducibility of audit records that contain no personal data.

Access audits and incidents

- Every read or write of a learner profile is logged with the caller, tenant, purpose and request ID.
- Access logs are reviewable, and anomalous access patterns raise alerts.
- An incident procedure covers detection, containment, notification of the host app and affected parties, and a post-incident review. Student content is not written to logs.

Child safety

- LAM must not infer or store sensitive attributes such as emotional or health conditions as facts.
- Escalation to a teacher or guardian is a recommendation to the host app, which owns the human workflow.
- Recommendations are validated so they cannot request unsafe content, and Qrie constrains what its tutor generates.

Proactive intervention triggers

Trigger	Signal	Response
Repeated failure	Three or more failures on a concept within a session	Diagnostic question or prerequisite check; stop rule if it continues
Long response time	Time well above the learner's own typical time for that difficulty	Offer a smaller step or a hint at the next level
Suspected guessing	Very fast responses plus low confidence or random pattern	Ask for reasoning; lower evidence weight
Hint dependency	Success only at assisted levels across several items	Return to independent attempts with scaffolding

Triggers are based on the learner's own baseline, and every inferred state is reported as probable with its confidence.

Metacognition and confidence calibration

- Confidence is collected by the host app where available and stored as unknown otherwise.
- Calibration is the gap between stated confidence and actual outcomes over a rolling window.
- Overconfidence (high confidence, wrong answers) prompts a reflective question. Underconfidence (low confidence, correct answers) prompts encouragement and evidence of progress.

Engagement and struggle signals

- Each signal has a definition, a permitted use and an uncertainty threshold.
- Fatigue, boredom and emotion are not inferred as facts. Signals such as long pauses may lower recommendation difficulty or offer a break, never label the learner.
- Signals below the confidence threshold do not change the learner state.

Multimodal evidence (later release)

Handwriting, voice and intermediate working steps are accepted only through new versioned input contracts. Each needs explicit consent, defined retention, confidence estimates for the extraction step, and a privacy review before release.